NMI Direct Connect

CardEase Integration Guide

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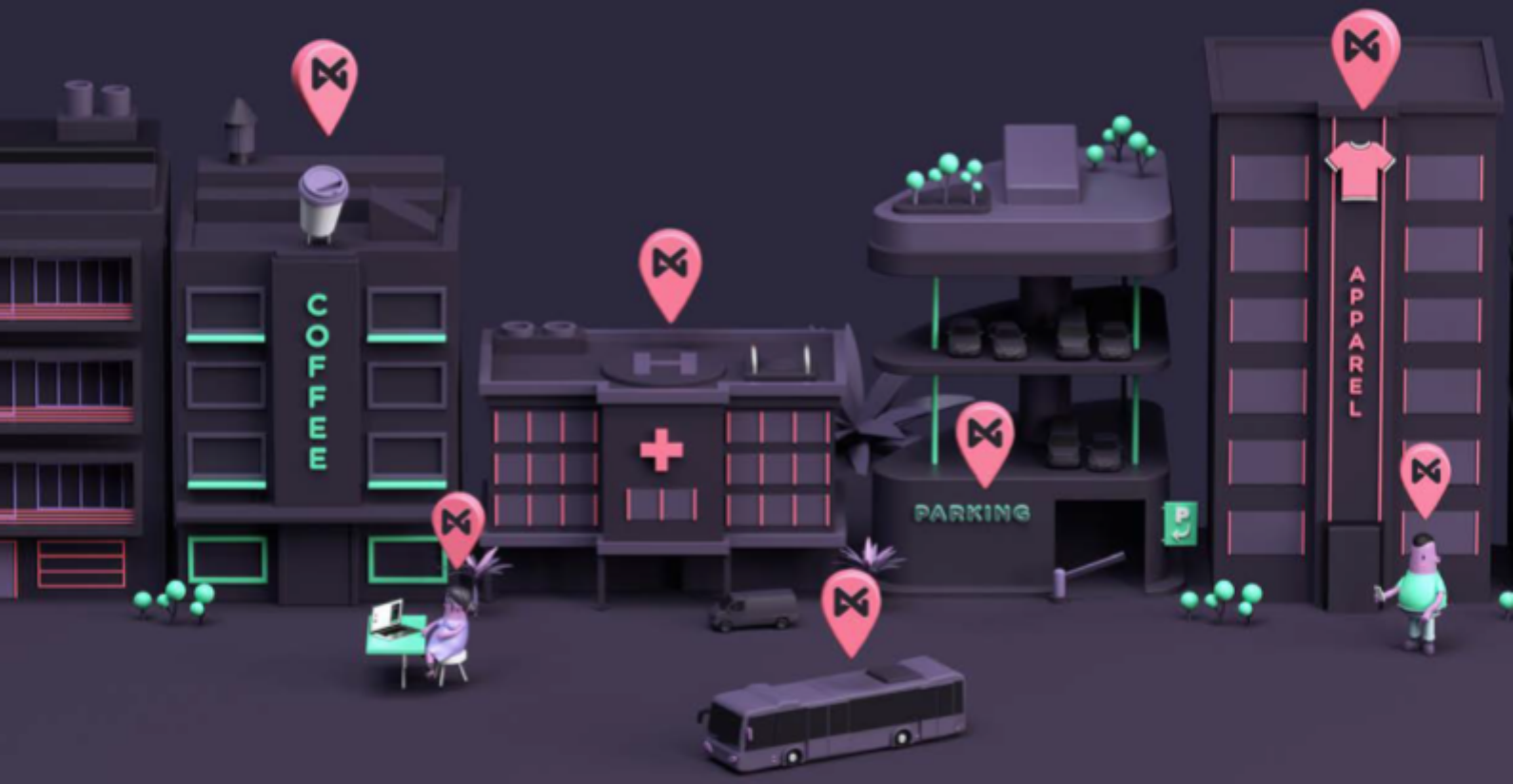

nmi

An integration guide describing the use of NMI Direct Connect cashless payment framework using NMI SDKs and protocols.

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Section 1 - Introduction

Who is NMI?

NMI provides payment enablement technology that empowers ISOs, ISVs, VARs and payment facilitators to offer branded payment gateway services without building or maintaining their technology. NMI went through an acquisition with Creditcall, an omnichannel payment gateway and EMV solutions provider, which was completed in March 2018.

For more information on the acquisition, please review the following blog post:
NMI Completes Acquisition of Creditcall

What is NMI Direct Connect?

NMI Direct Connect is a flexible client-server framework for processing credit and debit card transactions over an internet connection. It is based around NMI's CardEase platform and as a result, can handle manual (such as card-not-present), magnetic stripe and EMV details for several different transaction types such as 3-D Secure transactions, payment authorizations, recurring payments, and payment refunds.

Please ensure that you read and understand the attached terms and conditions.

This document describes in detail how to get started with using the NMI Direct Connect framework including:

- The Registration Process
- The Testing Framework
- The Web Front End
- The Transaction Types
- The Software Development Kits
- The Communication Protocol (with examples)

It should be noted that it is the responsibility of the integrator to ensure that all stages of the credit card transaction processing conform to the Payment Card Industry (PCI) guidelines.

EMV	Please note that the discussion of EMV transactions found within this document is only pertinent to the countries that support chip and pin transactions and as a result, will be highlighted. Those using NMI Direct Connect SDK outside of those countries can safely ignore this information.

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The NMI brand is respected as being synonymous with security and reliability. You may wish to include the NMI logo on your website using the following HTML:

```
<a href="http://www.nmi.com">
     <img
           alt="Secure Payments by NMI"
           border="0"

src="https://www.nmi.com/assets/uploads/site-uploads/nmi-badge.png"
     />
</a>
```


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Section 2 – Getting Started

Registration

Registration to use NMI Direct Connect can be performed by visiting this web site and completing the necessary form:
https://testwebmis.creditcall.com/

During the process of registration, it will be necessary to supply a valid email address and choose a password. An email will be sent to the supplied email address which will enable confirmation of registration.

Registration will provide details of an NMI Terminal ID and an NMI Transaction Key. It will be necessary to make a note of these as they will be required for integration, testing, and use of NMI Direct Connect.

Once registration is complete and all of the necessary details have been obtained, development can proceed.

Integration

Integration with NMI Direct Connect can be performed in two ways; by using a pre-existing SDK or by directly communicating with the NMI Direct Connect servers using the XML protocol. The use of the NMI Direct Connect SDKs or NMI Direct Connect protocol is described later in this document.

Whichever integration method is chosen, it is necessary to thoroughly test the resultant application to ensure that it works as expected.

Testing

During integration with the NMI Direct Connect platform, it is possible to make use of the NMI Direct Connect test server.

The majority of transactions submitted to the test server behave identically to those submitted to a live server, however, the corresponding bank transaction will not be performed against the card details used.

During testing, and after deployment is complete, it is possible to view all of the requests and the results of these within WebMIS.

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The NMI Direct Connect test server should not be used for load testing as Denial-Of-Service attack measures will cause the connecting IP addresses to be blocked.

WebMIS

To view requests made to the CardEase platform and their responses, it will be necessary to login to WebMIS using the username and password that were created at the time of registration.

WebMIS displays processed transactions and includes information such as:

- Terminal Information
- Approval Result
- Settlement State
- Masked Primary Account Number
- Transaction Totals

Knowledge Garden Articles

We recommend reviewing our Knowledge Garden (KG) articles as they include information which may be relevant to your integration. The Knowledge Garden includes general information, announcements, upcoming mandates as well as integration specific articles. It is also possible to subscribe to each subsection to ensure you receive an email notification for any new articles posted.

General/Announcements - https://support.nmi.com/hc/en-gb/categories/200058333

NMI Direct Connect Specific - https://support.nmi.com/hc/en-gb/sections/4409285373457-Direct-Connect

Completion/Go Live

After your integration has been finalized it must complete any applicable test cases listed in the NMI Solution Verification Program (NSVP) successfully. The NSVP is designed to ensure consistent quality of NMI Direct Connect solution integrations, and covers potential situations wherein the solution may not function in-line with NMI's expectations, resulting in loss of funds for the merchant.

Solutions cannot go live until they have been through this process and have had the results verified by NMI's Integration Team. This process must be completed for any initial solution deployments and releases that contain changes to the transaction flow. Please see the included NSVP documentation (`NMI Solution Verification Program.zip`) for more details.

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Section 3 - Integration

Communication

There are essentially two ways of communicating with the NMI Direct Connect servers; HTTPS connection or by using an established SDK made available by NMI.

Whichever communication method is chosen there are some common concepts and practices that should be noted. These are detailed below, followed by a more in-depth discussion of the possible transactions with the SDKs or NMI Direct Connect protocol.

Where possible a timeout should be specified for connecting and reading from the NMI Direct Connect servers. It is recommended that this is a minimum of 45 seconds to allow for the necessary bank authorization to occur. Connection keep-alive should be used where possible to reuse connections, reducing the negotiation overhead of multiple TLS connections.

TLS 1.2

Mirroring the PCI Security Council's (PCI SSC) guidance, all communication with our platform should be using TLS 1.2. The PCI SSC has indicated that if any additional vulnerabilities are found in TLS 1.1, they will mandate a move to TLS 1.2 or higher, a timeline that is out of our hands. Therefore, please ensure all integrations support TLS 1.2 at a minimum. For more information, including specific dependencies around the SDKs, please review the following article:
TLS 1.2 Support - NMI Direct Connect

Request/Response Model

The NMI Direct Connect framework operates around a request/response model to ensure that integrators can monitor the status of each transaction submitted to the CardEase platform.

By using one of the NMI Direct Connect SDKs the process can be performed by using Request, NMI Direct Connect Client and Response components:

- Request – the details of the request to send
- NMI Direct Connect Client – manages communication with the server(s)
- Response – the details of the response obtained

There is no need for the integrator to have an understanding of XML or to be concerned with message construction or parsing. All of these functions are performed by the SDK.

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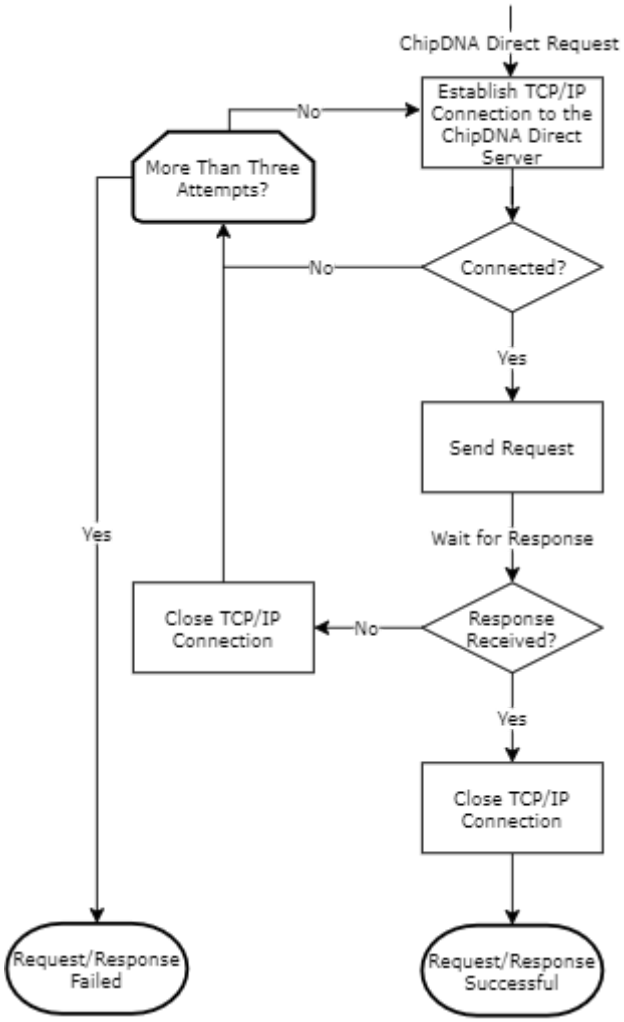
However, it should be noted that to use the NMI Direct Connect protocol outside of the available SDKs does require the user to construct a valid XML request which can then be submitted to the NMI Direct Connect server to obtain an XML response. This response will require parsing to extract the results of the transaction. Details of the XML requirements can be found later in this document.

The process flow is shown in the following diagram.



Testing

The test server can be found at the following URL:
`https://test.cardeasexml.com/generic.cex`

There are several special cases in which the test server can be used to simulate hard to replicate scenarios:

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Error Codes

To simulate the retrieval of a specific error code from the NMI Direct Connect server, the transaction amount should be substituted for the error code required. The amount unit should be minor.

EMV Issuer Script Data

To simulate the retrieval of EMV Issuer Script Data from the NMI Direct Connect server, the transaction amount should be specified as 400 (and the corresponding ICC tag 0x9f02). The amount unit should be minor.

Server URLs

Due to the distributed nature of NMI's Direct Connect processing network, several different data centers are used to handle transactions. NMI uses geographic routing via DNS so the test and production URLs route you to the nearest available data center.

When possible, we'd recommend using your ISP's DNS as this will ensure you're routed to the nearest NMI data center. When using 3rd party DNS providers, such as OpenDNS or Google DNS, it may result in your transactions being processed in a data center nearest to your DNS provider rather than the nearest data center to you.

Errors

Certain error cases should be managed in a way that minimizes downtime and inconvenience. The solution is dependent upon application and the type of error. A few of the suggestions are given below. All of the NMI Direct Connect errors can be found in appendix A.

Message	Description
PAN Missing	If possible, obtain the card details a second time
Card Reference Invalid	If possible obtain the original card details
Card Hash Invalid	If possible obtain the original card details
Temporarily Unavailable	If applicable, store the transaction for later processing

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A NMI Direct Connect integration should flag any errors that occur during a transaction for investigation. The solution should **stop** transacting if errors that have the potential to cause loss of funds occur until they have been investigated thoroughly, and the cause resolved. NMI cannot be held responsible for solutions that transact in error and will not be able to recover funds.

Test Cards

Overview

This article provides information on Card Not Present test PANs, where the card details are entered, rather than processed by a device, and physical test cards and where you can obtain them.

Physical test cards and test card PANs should not be used on the NMI production gateway and should be restricted to the test environment using a test account.

Please note that the test platform Special Amounts used to trigger events, such as a Decline or Voice Referral, should not be used in conjunction with PANs or physical test cards that are intended to invoke a specific response. For example, Special Amounts do not work in conjunction with the Test Card PANs for use with AVS noted below, as the AVS system requires an approval to generate the data.

Physical Test Cards

NMI is not a distributor of physical test cards, and as such we would recommend contacting your device distributor for advice on which cards would suit your use-case best.

On-Device Test Cards (Apple Pay™/Google Pay™)

On-Device test cards for use with Apple Pay™/Google Pay™ should be obtained through the relevant parties developer program, of which a sandbox account is provided and test cards able to be added to the respective digital wallet. For more information and the exact process of application please ensure you contact either relevant party.

Special notes. Attention

PANs programmed in our staging environment cannot be used with special amounts. Please, purchase another test card, often delaying sign-off of the solution (especially for transit integrations).

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Test Card PANs

Generic Card Scheme PANs

The NMI test platform will accept any valid PAN, however, if you don't have one available for use, it is possible to use one of the following available PANs:

Card Scheme	Card Number (PAN)
Amex	371111111111114
Diners	3001111111111116
Discover	6011111111111117
JCB	35281111111111119
Maestro	56410411111111113
Mastercard *(Credit)*	5411111111111115
Mastercard *(Debit)*	5511111111111114
Visa *(Credit)*	4532111111111112
Visa *(Debit)*	4751271111111118

** For expiry date, please choose any date in the future*
*** For CSC/CVV, please use the first 3 digits of the card number (PAN).*

Test Card PANs for use with AVS

The following test card PANs have addresses associated and can be used with either Direct Connect or Hosted Payment Page solutions to test Address Verification System response handling.

Card Scheme	Card Number (PAN)	CSC	Address	Postcode
Amex	376100000000004	3761	4 Amex Street, Southampton	SO31 6XY
Amex	341111597241002	1111	27 Broadway, New York	10004-1601

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Diners	36555500001111		Diners Club UK Ltd, 53 Diners Road, Salford	M5 3BH
Discover	6011000991300009	444	16 5TH ST SE, Washington DC	20003-1120
JCB	3561000000000005	356	5 JCB Street, Hereford	HR3 5TR
JCB	3566000020000410		Flat 10, 47 Park Street, London	W1K 7EB
Maestro	6761000000000006	676	6 Maestro Street, Exeter	EX16 7EF
Maestro	6333000023456788	888	1 Bd Victor, Paris, France	75015
Mastercard	5123450000000008	512	56 Gloucester Road, Glasgow	GL1 2US
Mastercard	5301250070000191	999	73 Whiteladies Road, Clifton, Bristol	BS8 2NT
Mastercard	5413330089600010	541	107 Central Park West, New York	10023
Mastercard	5413330089099049	541	1116 Amsterdam Ave, New York	10027
Mastercard	5761000000000008	576	8 Mastercard Street, Highbridge	TA6 4GA
Visa	4012000033330026	123	123 Fake Street, Sprinfield	SF1 234
Visa	4123450131003312	412	782 Northampton Lane, Hull	HL8 2UA
Visa	4111111111111111		28 Bishopgate Street, Sedgeford	PE36 4AW
Visa	4539791001730106	222	34 Broadway, New York	10004-1608
Visa	4761000000000001	476	1 Visa Street, Crewe	CW4 7NT
Visa	4761739001010010	476	84 Hudson St, New York	10013
Visa	4761739001012222	476	315 W 36th St, New York	10018
UK Maestro	6759111011100000	0	3 Wapping Road, Bristol	BS1 7YT
Int Maestro	6759184500000120768	777	1 Cornwallis Avenue, Clifton, Bristol	BS8 4PP
UK Maestro	675940111800103233	456	Flat 1, 10 High Street, Manchester	BL1 7TY
UK Maestro	676740055051020218	563	432 Main Road, Bath, UK	BA2 8UY

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** For expiry date, please choose any date in the future*

Test Card PANs for use with Transit Solutions

The following test card PANs can be used to test Funding PAN and PAR responses on the test platform. However, please note that:

- Whilst these can be used on an automatically boarded Terminal ID, they will not provide an FPAN/PAR response unless linked to a transit-enabled merchant account (MCC:4800). If this has not occurred, please reach out to our support team to action this for you.
- These values can only be used with a Direct Connect-based solution and not Payment SDK for Windows, Linux, Mobile, or Hosted Payment Page based solutions.
  - In a production transit environment, there is no guarantee that either PAR or FPAN value will be returned, as it relies on the acquirer and card issuer supporting the values.

PAN	Data Returned
374532345234322	PAR: NMI100130172780485734576140 52, FPAN: 376756541235167
376756541235167	PAR: NMI10013017278048573457614052
4023063263235350	PAR: NMI10013817305361759490045086
4023068595360560	PAR: NMI10013817305361759490045086
4038585432346230	PAR: NMI10015819260824338567883428
4038595360563650	PAR: NMI10015819260824338567883428
4532261249444950	PAR: NMI10017817225433759440047809, FPAN: 4761739001010010
5413330089600030	PAR: NMI10019662689345797536909754
5312270164858760	PAR: NMI10019662689345797536909754, FPAN: 5413330089600030
374245002771003	PAR: NMI10245818393736928367283545

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6510000000000210	PAR: NMI106197537283803179379426 84
6510000000000130	PAR: NMI106636790043369543612830 95
5413330089604110	FPAN: 5526133246657850
5514324356674320	FPAN: 5513324665785550
5534213456453450	FPAN: 5526133246657850
5734544352345230	FPAN: 5893586876094670
374245001751006	PAR: NMI104556712638693683676535 69
4761739001010010	PAR: NMI100178172254337594400478 09
4761739001010119	PAR: NMI100155196678293285678891 11

Special Amount

Overview

This article details the various special amounts that can be used on the staging platform to trigger an event, such as a decline or partial approval. This article is applicable to the following:

Supported Platforms

- Omni (Staging)
- CardEase (Staging)

Supported Solutions

- ChipDNA Windows & Linux
- ChipDNA Android & iOS
- ChipDNA Direct
- Hosted Payment Page

Need to Know

When working with a sandbox account there are constraints on the currency amounts you can use in test transactions. The amounts, by default, must be in the range of 0.25 to

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130.00 or the transaction will be declined with the error returned indicating the amount is too low/too high.

The special amounts noted below should only be used with a generic test card or PAN. If your integration utilizes a test card or PAN which is intended to generate a particular event you may not see the expected result if also using a special amount. For example, the amounts provided below do not work in conjunction with the AVS PANs noted in our Test Cards article, as that system requires an approval to generate the data.

Special Amounts

The following special amounts will trigger the corresponding result. Note that any other value will result in an approval, even when using an expired card.

Amount (Major Units)	Result
4.10	Approval
5.00	Decline
6.00	
10.27	Partial Approval (for 10.00)
14.29	SCA Fall Forward (AMEX)
14.30	SCA Fall Forward
14.31	SCA Online PIN Required
47.27	Partial Approval (for 40.00)
47.23	Voice Referral
48.03	Decline - Unable to connect to acquirer
48.04	Decline - Connection to acquirer timed out
48.05	Decline - Acquirer rejected request
48.06	Decline - Acquirer response invalid
49.01	Decline - Online PIN incorrect

** Minimum, maximum and special amounts all apply to the test platform only. These amounts do not create the same behavior on the live platform.*

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Section 4 – Requests and Responses

The following sections describe the different types of transactions that the NMI Direct Connect platform supports. This information should be used in conjunction with the NMI Direct Connect SDKs and NMI Direct Connect protocol to determine what information should be submitted and examined for different payment functions.

Authorization Transactions

An authorization request is a request for payment. This can include manual (such as card-not-present), magnetic stripe, recurring payment, and EMV transactions.

Manual payment requests must provide a PAN and expiry date as a minimum. Some card schemes also require the start date and issue number.

Magnetic stripe payment requests must provide the content of Track 2 as a minimum.

Recurring payment requests must provide a Card Reference and a Card Hash from a previous card transaction. The CardEase platform will use this information to determine the details of the card to be used. These details can then authorize a payment in the same way as a card-not-present authorization. **The Card Hash can and will change. It cannot be relied on for any operation other than for a repeat payment.**

Additional verification of the card information can be provided by supplying any combination of the address, card security code, and zip code/postcode. The authorization will not be accepted or declined based upon this information, however, it will provide the merchant with another level of verification that can be used to determine if the transaction should proceed. In cases where the transaction is declined, the authorization should be voided as to release the funds held against the account.

If 3-D Secure information is present due to the authorization request coming from an e-commerce transaction, this can be specified to provide the corresponding liability shift. If the 3-D Secure authentication fails due to the cardholder entering incorrect security data, the authorization should not be attempted. If it is, an error will be raised by either the NMI Direct Connect SDKs or the NMI Direct Connect server.

The response to the request will contain a result code (“Approved”, “Declined” or “VoiceReferralRequired”) and if the transaction has been approved an Authorization Code. If a voice referral is required the response will contain a voice referral telephone number.

If no errors occurred in processing the transaction the response will also contain details of the card that was processed. This might have been determined from manual (such as card-not-present), magnetic stripe, recurring payment or EMV ICC tags. This information

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will include PAN, expiry date, card scheme and depending upon the card scheme, start date and issue number.

If there was an error processing the transaction request then the response will contain one or more errors describing the problem. Errors can include items such as missing input, invalid input and transaction failures. For example, if the card has been previously blacklisted the result code would be “Declined” and the error “Card Blacklisted” would be returned. For a list of the possible errors see appendix A.

EMV Unlike manual (such as card-not-present) and magnetic stripe transactions more information is required to process an EMV authorization. All of the card detail has to be present in ICC tags which have been determined from the card. The number and content of the ICC tags that are required for authorizing a transaction in the EMV is dependent upon acquirer.

Each ICC tag must have a tag ID, a tag value type and a tag value.

The tag ID indicates what the content of the tag value is. It is a hexadecimal value that has been defined in the EMV specification. The content of the tag value can either be in an ASCII hex or string format and the tag value type is used to specify this.

The response to an EMV authorization request will contain the same information as the equivalent manual or magnetic stripe transaction. However, it will also contain several ICC tags that will need to be presented to the card or EMV kernel to confirm the authorization. Once done, another set of ICC tag information will be determined from the card which should be sent in a confirmation request so that settlement can take place.

Important Information

Please note that unless automatic confirmation is explicitly requested in the authorization transaction, all authorization transactions need to be followed by a confirmation transaction before settlement can take place. Automatic confirmation is not possible with EMV authorizations as they are a two-stage process.

Minimum Request Requirements:

- Auth message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request
- Amount to specify the amount of the transaction
- Card Details; one of:

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- Card Reference and Card Hash
- Manual: PAN and Expiry Date
- Magnetic Stripe: Track2
- EMV: ICC Type and ICC Tags

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:
  - Errors: Code and Message
  - Authorization and Result Code

Confirmation Transactions

A confirmation request is the second stage to an authorization request. It is used to confirm a previously approved authorization so that settlement can take place.

For a confirmation to take place, the CardEase Reference obtained from the authorization transaction should be presented in the request.

The specification of the confirmation amount is optional as it has already been sent in the authorization transaction. However, if required it can be specified to be less than the authorization amount in cases where the transaction amount decreases.

The response to the confirmation transaction is an acknowledgment of receiving the request and that the CardEase Reference is valid.

If there was an error processing the transaction request then the response will contain one or more errors describing the problem. Errors can include items such as missing input, invalid input and transaction failures. For example, if the CardEase Reference is invalid, the result code would be “Declined” and the error “CardEase Reference Invalid” would be returned. For a list of the possible errors see appendix A.

Important Information

Goods/services should only be provided to the end customer if an “Approved” result code is returned for an ***authorization and confirmation.*** This means both the authorization and confirmation have successfully been processed and therefore the funds for the transaction will be sent for settlement to the merchant. If an approval isn’t received for the confirmation, it means there was an issue with the confirmation and the transaction will not be sent for settlement. ***The merchant will not receive the funds for this transaction until you receive an “Approved” result code for the confirmation.*** We recommend waiting for an approval for both the authorization and the confirmation stage before providing goods/services to an end customer. Providing goods/services to an end

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customer after the authorization has approved but before the confirmation is approved will put the transaction at risk of not being funded to the merchant.

EMV In the case of an EMV request, all of the ICC tags that have changed since authorization are required in settlement. These typically include:

- Application cryptogram
- Cryptogram information data
- Application transaction counter
- Issuer application data
- Terminal verification results
- Transaction status information
- Issuer script results

Each ICC tag must have a tag ID, a tag value type and a tag value.

The tag ID indicates what the content of the tag value is. It is a hexadecimal value that has been defined in the EMV specification. The content of the tag value can either be in an ASCII hex or string format and the tag value type is used to specify this.

It should be noted that there is no ICC tag ID allocated for the Issuer Script Results. It is a requirement that the EMV Kernel/Terminal Integrator allocates their own. Therefore when it is a requirement to send the script results the tag ID must be set to “IssuerScriptResults”.

Minimum Request Requirements:

- Conf message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request
- CardEase Reference to specify the authorization transaction

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:
    - Errors: Code and Message
    - Result Code

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ICC Management Transactions

EMV

An ICC management request is used to perform additional functions that are required when using ICC capabilities. For example, it is possible to retrieve the sets of acquirer public keys.

To use the ICC Management request an ICC Management Function must be specified. At this time, the supported functions are:

- PublicKeyUpdate
  Retrieves a list of acquirer public keys.

Minimum Requirements:

- ICCManagement message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request
- ICC Management Function to specify the function to perform

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:
  - Errors: Code and Message
  - Result Code

Offline Transactions

Please note "Offline Transactions" are a legacy solution and new solutions should be using "Deferred Authorizations" unless told otherwise.

An offline request is used to settle transactions that have been performed on a terminal that has offline capability, such as an EMV or CAT terminal. In these situations, the transaction will have been approved offline and as a result, the data will have to be submitted post-authorization so that settlement can take place.

Magnetic stripe payment requests must provide the content of Track 2 as a minimum.

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EMV An offline EMV terminal will have provided certain ICC tags. These tags must be presented in an offline request for settlement. This list should contain the most recent values for the tags provided, such as the Application Cryptogram from the card and the Cryptogram Information Data.

Each ICC tag must have a tag ID, a tag value type and a tag value.

The tag ID indicates what the content of the tag value is. It is a hexadecimal value that has been defined in the EMV specification. The content of the tag value can either be in an ASCII hex or string format and the tag value type is used to specify this.

If there was an error processing the transaction request then the response will contain one or more errors describing the problem. Errors can include items such as missing input, invalid input and transaction failures. For example, if the PAN was not found the result code would be “Declined” and the error “PAN Missing” would be returned. For a list of the possible errors see appendix A.

Minimum Request Requirements:

- Offline message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request
- Amount to specify the amount of the transaction
- `OfflineDateTime` and `Format` to specify the date and time the offline transaction took place
- Offline card data; one of:
  - Track2
  - ICC Type and ICC Tags

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:
  - Errors: Code and Message
  - Result Code

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Deferred Authorization

Any given sale may require online authorization by the card, however, the ability to request this may not be available due a communication error between the solution and the platform. In circumstances where a response **has not** been received it may be possible to defer the authorization until such a time where communication has been restored. **If a decline response has been received for any other reason than a communication failure you should not attempt to defer the authorization**. It should also be noted that a Deferred Authorization request will automatically be marked for settlement if approved, and does not require manual confirmation. Automatic confirmation can be disabled by setting the `autoConfirm` element to `false` in the Deferred Auth request; the transaction will then require manual confirmation before being sent for settlement.

A Deferred Authorization should contain the exact same details as the original authorization attempt, but ensure the following elements have been changed/included:

Minimum Request Requirements:

- Message type changed to `DeferredAuth`
- `OfflineDateTime` and `Format` to specify the date and time the original authorization was attempted
- Include all original authorization details

Minimum Response Components:

- CardEase Reference, if response provided online, providing a unique reference to the transaction
- Local Date Time providing the date and time of the authorization
- Results; one of:
  - Errors: Code and Message
  - Result Code

Deferred Authorizations carry an inherent risk to the merchant, and do not ensure approval of a transaction. Before utilizing Deferred Authorizations with your solution you should:

- Discuss the process with NMI to ensure that all risks are understood and that the solution works as intended.
- Discuss the process with your processor to ensure that they are happy for Deferred Authorizations to be processed via a given merchant account, and that your solution incorporates any additional requirements (such as ceasing further attempts of a Deferred Authorization after a period of time has passed with no approval).
- Implement monetary and volume limits to ensure the merchant does not defer transactions they are unwilling to take the risk on, as **NMI cannot guarantee offline transactions will be approved**.

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- Perform daily reconciliation to ensure any and all deferred transactions are accounted for, and to monitor risk factors such as decline percentages, accumulated transaction volumes, or unacceptable values.

Please read this following Knowledge Garden article in full to further understand the deferred process: NMI Knowledge Garden - Deferred Authorization

Pre-Authorization Transactions

A pre-authorization request is a small authorization request to confirm that the card details are valid. This can include manual (such as card-not-present), magnetic stripe and EMV transactions. It is often used to confirm that a purchase can proceed before dispensing a product, after which the standard authorization and confirmation transactions can take place.

Manual payment requests must provide a PAN and expiry date as a minimum. Some card schemes also require the start date and issue number.

Magnetic stripe payment requests must provide the content of Track 2 as a minimum.

Recurring pre-authorization requests must provide a Card Reference and a Card Hash from a previous card transaction. The CardEase platform will use this information to determine the details of the card to be used. These details can then pre-authorize payment in the same way as a card-not-present pre-authorization.

Additional verification of the card information can be provided by supplying any combination of the address, card security code and, zip code/postcode. The authorization will not be accepted or declined based upon this information, however, it will provide the merchant with another level of verification that can be used to determine if the transaction should proceed.

The response to the request will contain a result code (“Approved”, “Declined” or “VoiceReferralRequired”) and if the transaction has been approved an Authorization Code. If a voice referral is required the response will contain a voice referral telephone number.

If no errors occurred in processing the transaction the response will also contain details of the card that was processed. This might have been determined from manual (such as card-not-present), magnetic stripe, recurring payment or EMV ICC tags. This information will include PAN, expiry date, card scheme and depending upon the card scheme, start date and issue number.

If there was an error processing the transaction request then the response will contain one or more errors describing the problem. Errors can include items such as missing input, invalid input and transaction failures. For example, if the card has been previously

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blacklisted the result code would be “Declined” and the error “Card Blacklisted” would be returned. For a list of the possible errors see appendix A.

EMV Unlike manual (such as card-not-present) and magnetic stripe transactions more information is required to process an EMV authorization. All of the card detail has to be present in ICC tags which have been determined from the card. The number and content of the ICC tags that are required for authorizing a transaction in the EMV is dependent upon acquirer.

Each ICC tag must have a tag ID, a tag value type and a tag value.

The tag ID indicates what the content of the tag value is. It is a hexadecimal value that has been defined in the EMV specification. The content of the tag value can either be in an ASCII hex or string format and the tag value type is used to specify this.

The response to an EMV authorization request will contain the same information as the equivalent manual or magnetic stripe transaction.

Minimum Request Requirements:

- PreAuth message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request
- Card Details; one of:
    - Manual: PAN and Expiry Date
    - Manual: Card Reference and Card Hash
    - Magnetic Stripe: Track2
    - EMV: ICC Type and ICC Tags

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:
    - Errors: Code and Message
    - Authorization and Result Code

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Refund Transactions

A refund request is used to refund a transaction that has previously been settled. A CardEase Reference obtained from the previous CardEase transaction must be used for this to take place.

If no amount is specified the previous transaction will be refunded in full, alternatively a smaller amount can be refunded. Additional refunds can be performed until the sum of all refunds against the specific CardEase Reference is equal to the original transaction amount.

The response to the request will contain a result code ("Approved", "Declined" or "VoiceReferralRequired"). If a voice referral is required the response will contain a voice referral telephone number.

If there was an error processing the transaction request then the response will contain one or more errors describing the problem. Errors can include items such as missing input, invalid input and transaction failures. For example, if the CardEase Reference was invalid the result code would be "Declined" and the error "CardEaseReferenceInvalid" would be returned. For a list of the possible errors see appendix A.

Minimum Request Requirements:

- Refund message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request
- CardEase Reference to specify the authorization request to refund

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:
    - Errors: Code and Message
    - Result Code

Test Transactions

A test request is used to test connectivity to NMI Direct Connect. This functionality should be used infrequently as excessive misuse may be interpreted as being a Denial-Of-Service attack causing the sender to be blocked from the NMI Direct Connect servers.

Minimum Requirements:

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- Test message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:
    - Errors: Code and Message
    - Result Code

Voice Referral Notification Transactions

A voice referral notification request is used to present NMI Direct Connect with the result of a voice referral. This should be performed to authorize or decline funds that have been previously marked as requiring a voice referral. Voice referrals are only applicable in an attended environment in which the merchant can call their acquiring bank with the telephone number provided in the original voice referral response.

For a voice referral notification to take place, a CardEase Reference obtained from the previous CardEase transaction must be specified, along with the result of the voice referral. If the voice referral has been approved an auth code will have been provided, this must be sent to NMI Direct Connect as the AuthCode.

If a connection to NMI Direct Connect is unavailable when trying to send a voice referral notification the voice referral notification should be stored for submission at a later time.

Minimum Requirements:

- VoiceReferralNotification message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request
- CardEase Reference to specify the authorization request to which the notification relates
- VoiceReferralResult to specify the result of the voice referral request

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:

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- Errors: Code and Message
- Result Code

Void Transactions

A void request is used to void the original transaction that has been authorized but not settled. This should be performed to release funds that have been reserved against an account when the transaction will no longer take place.

In order for a void to take place, a CardEase Reference obtained from the previous CardEase transaction must be specified, along with a reason for the void.

If a connection to NMI Direct Connect is unavailable when trying to void a transaction the void should be stored for submission at a later time.

It should be noted that batch voids are not supported by NMI Direct Connect. The CardEase systems provide a single interface to many different methods of authorizing payments and many of these require the voiding failed transactions as soon as possible. This can be to release funds held against an account or due to the different processes that acquirers have towards settlement. The use of batch void would mean that there would be a risk of settlement on a transaction that should have been voided. The resultant refund would incur extra cost to the client and the acquirer as well as causing customer confusion.

Minimum Requirements:

- Void message type
- Terminal ID to specify the terminal issuing the request
- Transaction Key to specify the transaction validating the request
- Software Name to specify the software making the request
- Software Version to specify the software making the request
- CardEase Reference to specify the authorization request to void
- Void Reason to specify the reason the void was being performed

Minimum Response Components:

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results; one of:
  - Errors: Code and Message
  - Result Code

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Partial Reversals

In certain situations, the amount to be settled will be lower than the original authorized amount. In this scenario, where supported, a partial reversal may be performed for the difference between the authorized amount and the confirmed amount. This should be indicated in the authorization request by setting the amount type equal to “Estimate”.

Funding PAN (FPAN) / Payee Account Reference (PAR)

What Is FPAN / PAR?

Funding PAN (FPAN)

When a physical card is added into a digital wallet service, the PAN is tokenized into a Device PAN (DPAN). This DPAN is unique to that device, and results in a single physical card having a different DPAN across multiple devices (the DPAN may also change if the same card is re-added to the Digital Wallet on the same device). The physical card PAN is referred to as the Funding PAN (FPAN), the value of which is tokenized by the NMI Gateway, and can be used to refer back to the original card, or in some cases, used to submit further charges to that card where not possible with the DPAN.

The FPAN Reference and Hash may also allow CNP transactions using this value, such as for Debt Recovery.

Payee Account Reference (PAR)

A Payee Account Reference is a unique reference for a single card that can be used to identify and refer back to a cardholder when using the physical card or DPAN via a Digital Wallet payment. It should be noted that if a cardholder has two cards issued by the same card issuer, each card will have a unique PAR value.

The PAR value can be used to link a transaction performed with a DPAN to the FPAN, or vice versa, by comparing the value returned to the originating card or device.

When Is the FPAN or PAR Value Returned?

NMI does not control whether your integration will receive an FPAN or PAR in a response, and can be dependant on:

- The Merchant Category Code for the Merchant Account processing the transaction.
- Whether the card scheme supports returning of the FPAN or PAR value.
- Whether your processor supports returning FPAN or PAR value.

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- Whether NMI's integration for the processor supports returning the FPAN or PAR value.

FPAN details are returned in the response FundingCard element and the PAR value is returned in the response CardDetails element.

Please see appendix B for a list of PANs that can be used on the test platform to obtain a FPAN and/or PAR response.

Soft Decline SCA Indicator

What Is The SCA Indicator?

Card issuers in the European Union and European Economic Area (EU/EEA) will soon be required to comply with updated PSD2, Strong Customer Authentication (SCA) regulatory requirements. For contactless transactions, this requires two-factor authentication when 150 Euros or 5 consecutive contactless transactions have been completed since the card was last authenticated. A decline under these conditions is known as a soft decline and should be re-attempted.

Depending on the card scheme, the issuer may request the card be further authenticated by re-submitting the transaction with Online PIN and/or as a Contact transaction.

SCA Fall Forward Flow

Fall Forward To Contact:

1. Contactless authentication attempted.
2. Transaction declined with **`FallForwardToContact`** indicator in response with value of true (boolean).
3. Retry authentication via an EMV contact-only transaction.
    a. Include the original platform transaction reference in a TransactionLinks block with a type of **`FallForwardToContactInitiator`**.

Online PIN Required:

1. Contactless authentication attempted.
2. Transaction declined with **`OnlinePinRequired`** indicator in response with value of true (boolean).
3. Prompt for PIN entry and re-submit the authentication.
    a. The PIN block should be stored as part of **`ExtendedProperties`**.

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b. Include the original platform transaction reference in a TransactionLinks block with a type of `OnlinePinRequiredInitiator`.

Minimum Response Components

- CardEase Reference providing a unique reference to the transaction
- Local Date Time providing the date and time of the transaction
- Results, one or more of depending on card scheme and issuer:
  - Acquirer Response Code to categorize the fall forward transaction type
  - Fall Forward To Contact indicator
  - Online PIN Required indicator

Credential-On-File - Cardholder Initiated and Merchant Initiated Transactions

What Is Credential-on-File?

A Credential-on-File (COF), or Card-on-File transaction is where a cardholder authorizes a merchant to store and subsequently charge their card at a later date (e.g. tokenization). Credential-on-File transactions can be initiated by either the cardholder or the merchant.

Cardholder initiated transactions are the cardholder requesting the merchant to initiate a transaction. This could be paying for goods in a store, over the phone or online.

Merchant initiated transactions are the merchant processing a transaction as a result of an agreement with the cardholder such as debt recovery.

Any tokenized transactions using the card reference and card hash (e.g. the token) need to abide by the implementation details below.

Credential-on-File Flow

The below applies for the following transaction types:

- Auth
- Pre-Auth

Initial Transaction - First Store

1. Perform a transaction using the raw card details (EMV data/full pan) and flag who initiated the transaction (the merchant or the cardholder). This indicates the transaction as being a COF initiated transaction that will be used for COF later. This transaction is known as the 'first store'. See example 1.

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   a. Ensure to flag if it was the `CardHolder` or `Merchant` who initiated the transaction using the `TransactionInitiatedBy` property.
   b. Include the `TransactionReason` if possible.

2. The card reference and card hash (token) are returned in the transaction response from our platform. This can be used at a later date to perform a subsequent, tokenized transaction flagged as COF.
   a. Ensure to store the card reference, card hash, and the original CardEase Reference as they'll be needed when performing COF transactions.

Please refer to the API docs for a full list of valid property values.

Subsequent Transactions

1. Perform a transaction using the card reference and card hash (token). See example 2.
   a. Ensure to include the original CardEase Reference within the `CardEaseReference` property from the initial 'first store' transaction performed.
   b. Ensure to flag if it was the `CardHolder` or `Merchant` who initiated the transaction using the `TransactionInitiatedBy` property.
      i. Include the `TransactionReason` if possible.

Please refer to the API docs for a full list of valid property values.

Overriding Default First Store Behaviour

The presence of Card Data (PAN/Track 2) will, by default, determine if a transaction is a first store transaction. In the absence of Card Data, a tokenized transaction (using Card Hash and Card Reference) will default to indicate that the transaction is a subsequent transaction.

If you are using your own card store and sending card details in each card transaction, instead of a token, then it is necessary to include the FirstStore flag to override the default.

C# Example 1 - Initial Transaction (First Store)

```
Request request = new Request();

 request.RequestType = RequestType.Auth;
 request.Amount = "123";
 request.SoftwareName = "SoftwareName";
 request.SoftwareVersion = "SoftwareVersion";
 request.TerminalID = "********";
 request.TransactionKey = "****************";

 request.PAN = "***************";
```


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```
 request.ExpiryDate = "****";

 TransactionInitiatedBy transactionInitiatedBy =
TransactionInitiatedBy.CardHolder;
 TransactionReason transactionReason =  TransactionReason.Empty;

 CredentialOnFile cof = new CredentialOnFile(
 transactionInitiatedBy,
 transactionReason
 );

 request.CredentialOnFile = cof;

 /*Setup the client*/
 Client client = new Client();
 client.AddServerURL("https://test.cardeasexml.com/generic.cex", 45000);
 client.Request = request;

 try
 {
 /*Process the request*/
 client.ProcessRequest();
 }

 Response response = client.Response;
 string cardHash = response.CardHash; /*Storing Card Hash for subsequent
transactions*/

 string cardReference = response.CardReference; /*Storing Card Reference for
subsequent transactions*/

 string initialcardEaseReference = response.CardEaseReference; /*Storing
CardEase Reference for subsequent transactions*/
```

C# Example 2 - Subsequent Transactions

```
Request cofRequest = new Request();

 cofRequest.RequestType = RequestType.Auth;
 cofRequest.Amount = "123";
 cofRequest.SoftwareName = "SoftwareName";
 cofRequest.SoftwareVersion = "SoftwareVersion";
 cofRequest.TerminalID = "********";
 cofRequest.TransactionKey = "****************";

 cofRequest.CardHash = cardHash; /*Stored Card Hash from initial transaction*/
 cofRequest.CardReference = cardReference; /*Stored Card Reference from initial
transaction*/

 TransactionInitiatedBy transactionInitiatedBy =
TransactionInitiatedBy.Merchant;
 TransactionReason transactionReason = TransactionReason.Resubmission;
```


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```
CredentialOnFile cof = new CredentialOnFile(
transactionInitiatedBy,
transactionReason,
initialcardEaseReference /*Stored CardEase Reference from initial transaction*/
);

request.CredentialOnFile = cof;

client.Request = request;

try
{
/*Process the request*/
  client.ProcessRequest();
}
```

The above information and examples are also available on our Knowledge Garden.

Incremental Authorizations

What are Incremental Authorizations?

Incremental Authorizations allow an integrator to increase the amount of the initial authorisation by an incremented amount prior to the transaction being confirmed/captured. In doing so, the initial authorization and each increment thereafter are authorized separately, and after Capture, the transaction appears as one entry on the customer's statement.

Initial Authorization

Incremental Authorizations are performed by sending an initial authorization, followed by subsequent, incremented transactions. Both an Initial Authorization and Subsequent transactions should be sent using the CredentialOnFile field, and can be utilized for both card present and card not present requests (example below).

```
Request request = new()
{
      ...
      CredentialOnFile = new(
            initiatedBy: TransactionInitiatedBy.CardHolder,
            reason: TransactionReason.Incremental,
            firstStore: TransactionFirstStore.True
      ),
      ...
}
```


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Subsequent Authorizations

Follow up subsequent transactions similarly utilize the CredentialOnFile field, ensuring a reference to the initial store is submitted, and the initiatedBy, reason, and firstStore fields set appropriately (example below). Subsequent authorizations should be submitted as card not present transactions, please use the card tokens (CardHash and CardReference) from the response of the first store in lieu of card data. The amount specified in subsequent authorizations should not be cumulative but the amount to be incremented by.

```
Request request = new()
{
      ...
      CredentialOnFile = new(
            initiatedBy: TransactionInitiatedBy.Merchant,
            reason: TransactionReason.Incremental,
            firstStore: TransactionFirstStore.False,
            cardEaseReference: $"{firstTransaction-GUID}"
      ),
      ...
}
```

Void

It is possible to cancel/void both a first store incremental auth and any or all of the subsequent incremental authorizations. If the entire transaction, including any incremental authorizations associated with the first store, needs to be voided, initiate a normal void request against the first store authorization. Alternatively, simply send a normal void request against an incremental authorization. These void requests follow normal procedures, and as a result, a CredentialOnFile field is not expected in these requests.

Final Confirmation

Once all of the subsequent incremental authorizations have been submitted and the final amount is known the transaction must be captured. To do so, a capture request must be sent against the initial authorization (first store), if no amount is included in the request, the platform will automatically capture the full amount automatically. These confirmation requests follow normal procedure, and as a result, a CredentialOnFile field is not expected in these requests.

Please note that:

- The confirmation must be equal or lower than the total amount;

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- Once the transaction is confirmed no further subsequent incremental authorizations are permitted for the transaction.

Incremental Authorization Example

Example Initial/First Store Auth (first store incremental)

```
Request request = new()
{
      Amount = $"{Amount}",
      CredentialOnFile = new(
            initiatedBy: TransactionInitiatedBy.CardHolder,
            reason: TransactionReason.Incremental,
            firstStore: TransactionFirstStore.True
      ),
      ICCType = "EMV",
      ICCTags = new()
      {
            new("0x57", "5413330089000013D0712201087490938F"),
            new("0x5a", "5413330089000013"),
            new("0x5f24", "071231"),
            new("0x5f34", "00"),
            new("0x9f02", "000000000123"),
            new("0x9f03", "000000000000"),
            new("0x9f26", "095BDFF410C27326"),
            new("0x82", "1800"),
            new("0x9f36", "0003"),
            new("0x9f37", "28BB7FE5"),
            new("0x95", "800000E000"),
            new("0x9c", "00"),
            new("0x9f10", "020000000000"),
            new("0x9f06", "A0000000041010"),
            new("0x9f09", "0002"),
            new("0x9f27", "80"),
            new("0x9f34", "410302"),
            new("0x9f35", "24"),
            new("0x9b", "6800"),
            new("0x4f", "A0000000041010"),
            new("0x9f08", "0002"),
            new("0x9f07", "FF00"),
            new("0x5f28", "0056"),
            new("0x9f0d", "F040642000"),
            new("0x9f0e", "0010880000"),
            new("0x9f0f", "F0E064F800"),
            new("0x9f33", "6098C8"),
            new("0x9a", "060424"),
            new("0x5f20", ICCTagValueType.String, "REQ01 MC"),
      },
      RequestType = RequestType.Auth,
      SoftwareName = $"{SoftwareName}",
      SoftwareVersion = $"{SoftwareVersion}",
      TerminalID = $"{TerminalID}",
      TransactionKey = $"{TransactionKey}",
```


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```
};
```

Request Example of Subsequent Auth

```
Request request = new()
{
      Amount = $"{Amount}",
      CardHash = $"{firstTransaction-CardHash}",
      CardReference = $"{firstTransaction-CardReference}",
      CredentialOnFile = new(
            initiatedBy: TransactionInitiatedBy.Merchant,
            reason: TransactionReason.Incremental,
            firstStore: TransactionFirstStore.False,
            cardEaseReference: $"{firstTransaction-GUID}"
      ),
      ManualType = "cnp",
      RequestType = RequestType.Auth,
      SoftwareName = $"{SoftwareName}",
      SoftwareVersion = $"{SoftwareVersion}",
      TerminalID = $"{TerminalID}",
      TransactionKey = $"{TransactionKey}",
};
```

Example Final Confirm

```
Request request = new()
{
      CardEaseReference = $"{firstTransaction-GUID}",
      RequestType = RequestType.Conf,
      SoftwareName = $"{SoftwareName}",
      SoftwareVersion = $"{SoftwareVersion}",
      TerminalID = $"{TerminalID}",
      TransactionKey = $"{TransactionKey}",
};
```

Debt Repayment

What Is Debt Repayment?

A Debt Repayment transaction is one where a consumer is repaying debt to a merchant. If the transaction is a form of debt repayment, this should be flagged using the `DebtRepayment` property. By default, this value is set to `false` if not sent.

Debt Repayment is only permitted under certain Merchant Category Codes.

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3D Secure

What is 3D Secure

3D Secure (3DS) is a security protocol used to authenticate users. This provides an extra layer of protection for payment card transactions in card-not-present scenarios. It was designed to allow a cardholder to authenticate its identity to prevent payment fraud, stymie unauthorized transactions, and reduce chargebacks.

C# Example - Successful 3DS v2 transaction

```
Request request = new Request();

 request.RequestType = RequestType.Auth;
 request.Amount = "123";
 request.SoftwareName = "SoftwareName";
 request.SoftwareVersion = "SoftwareVersion";
 request.TerminalID = "********";
 request.TransactionKey = "****************";

 request.PAN = "***************";
 request.ExpiryDate = "****";

 request.ThreeDSecureCardHolderEnrolled = ThreeDSecureCardHolderEnrolled.Yes;
 request.ThreeDSecureTransactionStatus =
ThreeDSecureTransactionStatus.Successful;
 request.ThreeDSecureECI = "**";
 request.ThreeDSecureIAV = "****************";
 request.ThreeDSecureIAVAlgorithm = "*";

 request.ThreeDSecureVersion = "***********";
 request.ThreeDSecureServerTransactionId =
"********-****-****-****************";

request.ThreeDSecureDirectoryServerTransactionId =
"********-****-****-****************"

 /*Setup the client*/
 Client client = new Client();
 client.AddServerURL("https://test.cardeasexml.com/generic.cex", 45000);
 client.Request = request;

 try
 {
 /*Process the request*/
 client.ProcessRequest();
 }
```


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3DS2.2. Exemption Flagging

One key difference between 3DS 2.1 and 2.2 is the ability to support exemptions.

3DS 2.2 also allows merchants to request exemptions through their acquirer. This includes the merchant or payment service provider can apply Transaction Risk Analysis (TRA) and use this data to request for a low-risk exemption. It also allows merchants to request an exemption as a Trusted merchant.

Exemption flagging is supported from version 5.3.0 of the CardEaseXML DirectClient.

3DS2.2 Example - SCA Exemption Flagging

```
Request request = new()
{
      Amount = "1000",
      CurrencyCode = "GBP",
      ExpiryDate = "2312",
      ExpiryDateFormat = "yyMM",
      PAN = "4444444444444448",
      CSC = "123",
      RequestType = RequestType.Auth,
      ScaExemption = ScaExemption.LowValue,
      TerminalID = "99900001",
      TransactionKey = "ABCDEF1234",
    UserReference = "Test1",
};
```

Terminal Capabilities

What are Terminal Properties?

It is important that an integration provides the gateway with a full understanding of its capabilities so that the gateway can make correct decisions about how to process transactions, and pass relevant information to the target acquirer / processor.

Including this information is optional for existing integrations and certifications, however newer integrations should populate this data.

C# Excerpt

```
TerminalCapabilities terminalCapabilities = new TerminalCapabilities();
Parameters parameters = new Parameters();
parameters.Add(ParameterKeys.AttendedType,ParameterValues.AttendedTypeAttended);
parameters.Add(ParameterKeys.DeviceType, ParameterValues.DeviceTypeConsumer);
```


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```
parameters.Add(ParameterKeys.MobilePos, ParameterValues.True);
parameters.Add(ParameterKeys.Premises, ParameterValues.PremisesTypeOnPremises);
parameters.Add(ParameterKeys.Keyed, ParameterValues.True);
parameters.Add(ParameterKeys.Magstripe, ParameterValues.True);
parameters.Add(ParameterKeys.ContactEmv, ParameterValues.False);
parameters.Add(ParameterKeys.ContactlessEmv, ParameterValues.True);
parameters.Add(ParameterKeys.Visa, ParameterValues.True);
parameters.Add(ParameterKeys.Mastercard, ParameterValues.True);
parameters.Add(ParameterKeys.Amex, ParameterValues.False);
terminalCapabilities.AddCardInputCapabilities(parameters);
terminalCapabilities.AddCardholderActivatedTerminalCapabilities(parameters);
terminalCapabilities.AddGenericCapabilities(parameters);

Request request = new()
{
      ...
      TerminalCapabilities = terminalCapabilities,
      ...

};
```


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Section 5 – SDKs

Overview

Using an SDK developed by NMI allows for quicker integration as it is using a codebase that has been tested to conform to the NMI Direct Connect protocol. There is no requirement for the integrator to be concerned with the construction and parsing of XML data or the communication with the NMI Direct Connect servers.

The SDK discussion in this document is not intended to be a detailed description of how to use the language-specific components of the SDKs, however, several common features are presented for clarity. For more detailed documentation that presents the language-specific components, the SDK content should be consulted.

At this time client SDKs are available for the following development platforms:

- .NET 2.0
- Android
- Java 1.8
- Objective-C
- PHP

Each of the SDKs includes three core components:

- Development library (DLL, JAR, PHP, etc.)
- Documentation (CHM, HTML etc.)
- Examples (C#, Java, PHP etc.)

If you require a specific SDK that is not listed, please feel free to contact us so that we can consider inclusion for later releases.

Features

All of the NMI Direct Connect SDKs include support for the following features:

- All available request types (Auth, Conf, Refund etc.)
- Transparent construction and parsing of XML data
- Multiple server URL specification for redundancy
- HTTPS Proxy specification and authentication

Where language constructs allow, additional support is included for generic types and enumeration.

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Validation of data

The NMI Direct Connect SDKs perform only basic validation of request and response data. The reason for this is primarily for performance and to avoid unnecessary duplication of functionality since the NMI Direct Connect servers also validate all input.

When submitting a request the SDKs will ensure that all necessary data components are present, such as request type and terminal ID. Likewise, when receiving a response, the data will be validated to ensure that all required components are present such as CardEase Reference and result code. The content of the request and response data will not be validated for conformity for standards.

If validation of request data is required before submission to the NMI Direct Connect servers due to an unreliable source (such as manual entry) this should be performed at a stage prior to constructing the request data. Consideration should be given to:

- Trimming of white space from the start and end of input
- Confirmation input does not contain empty strings
- Performing a Luhn check to ensure PANs are valid
- Checking that the dates conform to the date format
- Validation that transaction amounts fall within valid ranges

Proxy Support

All of the SDKs have support for proxy hosts and proxy authentication. This allows for users inside corporate firewalls to connect to the NMI Direct Connect servers through a company proxy server. When using such a facility ensure that the proxy hostname and port are set, and if required the proxy username and password.

Dependencies

Please refer to the following article for our SDK dependencies.

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Appendices

A - Error Codes

The possible error codes that can be returned from a NMI Direct Connect request.

Error Code	Message	Description
1001	Expired Card	The specified card in the request has expired
1002	Pre Valid Card	The specified card in the request is not yet effective
1003	Card Scheme Not Supported	The specified card scheme in the request is not supported
1004	Card Usage Exceeded	The specified card usage in the request has been exceeded
1005	Card Banned	The specified card in the request has been banned
1006	Not Allowed	The specified transaction in the request is not allowed
1200	PAN Missing	The request does not contain a PAN
1201	PAN Invalid	The specified PAN in the request is invalid
1202	PAN Too Long	The specified PAN in the request is too long
1203	PAN Too Short	The specified PAN in the request is too short
1204	PAN Fails Luhn Check	The specified PAN in the request fails the Luhn check
1210	Expiry Date Missing	The request does not contain an expiry date
1211	Expiry Date Invalid	The specified expiry date in the request is invalid
1220	Start Date Missing	The request does not contain a start date.
1221	Start Date Invalid	The specified start date in the request is invalid
1230	Issue No Missing	The request does not contain an issue number
1231	Issue No Invalid	The specified issue number in the request is invalid
1235	Card Reference Invalid	The specified card reference in the request is not valid
1236	Card Hash Invalid	The specified card hash in the request is not valid
1240	Amount Missing	The request does not contain an amount
1241	Amount Invalid	The specified amount in the request is invalid

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1242	Amount Too Small	The specified amount in the request is too small
1243	Amount Too Large	The specified amount in the request is too large
1250	Message Type Missing	The request does not contain a message type
1251	Message Type Invalid	The specified message type in the request is invalid
2001**	Terminal ID Missing	The request does not contain a terminal ID**
2002	Terminal ID Unknown	The specified terminal ID is unknown, missing, invalid or disabled
2003**	Terminal ID Invalid	The specified terminal ID in the request is invalid**
2004**	Terminal ID Disabled	The specified terminal ID in the request is disabled**
2005	Terminal Usage Exceeded	The specified terminal ID usage in the request has been exceeded
2021	Transaction Key Missing	The specified transaction key in the request is missing
2022**	Transaction Key Invalid	The specified transaction key in the request is invalid**
2023**	Transaction Key Incorrect	The specified transaction key in the request is incorrect**
2100	CardEase Reference Missing	The request does not contain a CardEase Reference
2101	CardEase Reference Invalid	The specified CardEase Reference in the request is invalid
2110	Card Details Unavailable	The card details referenced by the Card Reference and Card Hash are unavailable
2111	Card Details Not Found	The card details referenced by the Card Reference and Card Hash could not be found
2200	Transaction Not Found	The specified transaction in the request was not found
2201	Transaction Already Settled	The specified transaction in the request has already been settled
2202	Transaction Already Voided	The specified transaction in the request has already been voided
2203	Transaction Already Refunded	The transaction has already been refunded in full
2204	Transaction Originally Declined	The specified transaction in the request was originally declined
7000	Temporarily Unavailable	The CardEase platform is temporarily unavailable
8001	Invalid XML Request	The request XML is invalid

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8002	Invalid Message Type	The specified request type is invalid
8003	XML Element Missing	The request does not contain all of the expected XML elements
8004	Invalid Data	An invalid piece of information was sent in the request
8005	XML Decryption Error	It is not possible to decrypt the XML

** This error code was valid until March 15, 2023. After this date, the error is consolidated under error code 2002 TerminalIDUnknown.

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B - Test Card PANs

For a list of available test card PANs which can be used to trigger a variety of responses, please see our Test Cards Knowledge Garden article.

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Glossary

Term	Description
Acquirer	An acquirer is a company offering debit and credit card acceptance services for merchants.
Authorization Transaction	An authorization transaction is a request for payment. This can include manual (such as card-not-present), magnetic stripe, recurring payment, and EMV transactions.
Capture	Capture is the point at which the card details are obtained. This may be from a variety of sources such as a magstripe reader or by physical inspection of the card.
Card Not Present	A transaction where neither the card nor the cardholder is present at the point of sale. For example, orders over the Internet, by mail, telephone or fax.
Confirmation Transaction	A confirmation request is the second stage to an authorization request. It is used to confirm the authorization obtained so that settlement can take place.
CSC	The Card Security Code printed on credit and debit cards. It can be used as an additional check to allow the merchant to decide whether to proceed with the transaction. Also known as CVV, CV2, CVV2, CVC, CVC2, and CID.
EMV	Europay Mastercard Visa standard for the interoperation for IC cards and IC capable POS terminals for authenticating credit and debit card payments. Also known as IC Credit and Chip and PIN.
EMV Transaction	An EMV transaction is one in which the card details have been obtained by electronic communication with the microprocessor on the card. The card details will be communicated in the form of ICC Tags.
Keyed Transaction	A keyed transaction is one in which the card details have been obtained by a manual means such as physical inspection of the card, or over the telephone and the details are entered by hand.
ICC	An Integrated Circuit Card (ICC) is a card with an embedded microprocessor. These are used for EMV transactions.
ICC Tag	An ICC Tag is a piece of information that can be sent to and received from an ICC. These tags have an id, type, and value. These are used for EMV transactions.
Issuer	The issuer of the credit card. For example, Bank Of America may be the issuer of a Visa card.
Luhn Check	A simple checksum formula for validating credit card numbers. It helps protect against accidental input error. Also known as the "modulus 10" or "mod 10" algorithm.
Magnetic Stripe	A magnetic stripe transaction is one in which the card details have been obtained by

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Transaction	passing the card through a magnetic stripe reader. The magnetic stripe (or magstripe) contains 3 tracks, track 2 is mandatory for authorizations.
Manual Transaction	A manual transaction is one in which the card details have been obtained by a manual means such as physical inspection of the card, or over the telephone and the details are entered by hand.
Merchant	The Merchant is the entity which is requesting funds from the cardholder's account via the Acquirer and Issuer.
Offline Transaction	An offline transaction is used to settle CAT or EMV transactions that have been performed on a terminal that has offline capability. In these situations, the transaction will be approved offline and as a result, the data will have to be submitted post-authorization so that settlement can take place.
PAN	The Primary Account Number (PAN) is the number that is found on the front of a credit card and is used to uniquely identify an account. Also known as a Personal Account Number.
Pre-Authorization Transaction	A pre-authorization request is a small authorization request to confirm that the card details are valid. This can include manual (such as card-not-present), magnetic stripe and EMV transactions.
Refund Transaction	A refund transaction is used to refund a transaction that has previously been settled.
Settlement	Settlement is the point at which the authorized transaction amount is requested from the Issuer to pay the Merchant.
Void Transaction	A void transaction is used to void a transaction that has been authorized but not settled.

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Terms and Conditions

1 Definitions

1.1 “Agreement” means this agreement.

1.2 “Network Merchants IPR” means any and all current and future registered and unregistered trademarks and service marks, patents, registered and unregistered designs, applications for any of the foregoing, trade names, copyrights, inventions, know how, protocols and other forms of intellectual property.

1.3 “Network Merchants Offering” means the Site, the Site Content and/or the Services in whole or in part.

1.4 “Person” means any person, firm, company, corporation, government, state or agency of a state or any associated, subsidiary or holding company or partnership (whether or not having separate legal personality) or one or more of the foregoing.

1.5 “Program Concepts” means the techniques and ideas embodied and expressed in the Network Merchants IPR.

1.6 “Services” means any current or future service(s) provided to You by Network Merchants including but not limited to access to the Site and/or the Site Content.

1.7 “Site” means the Network Merchants CeXML website.

1.8 "Site Content" means all the text, images, marks, logos and other content of the Site including but not limited to the Network Merchants IPR, Support Materials and Program Concepts.

1.9 “Support Materials” means any human-readable or machine-readable program listings, executable code, flow charts, logic diagrams, input and output forms, manuals, specifications, test cards, instructions, and other materials, and any copies of any of the foregoing in any medium related to the Network Merchants IPR and delivered to You in accordance with this Agreement.

1.10 “Terms” means the terms and conditions set out in this Agreement and any operating rules, regulations, policies, and procedures that may be notified to you by Network Merchants either in writing, verbally or via the Site.

1.11 “You” means the Person agreeing to comply with the Terms in return for accessing the Services.

2 General

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2.1 This Agreement governs the provision of the Network Merchants Offering to You on the Terms set out herein.

2.2 Except in the case of negligent or fraudulent misstatement this Agreement constitutes the entire agreement between Network Merchants and You for your access to and/or use of the Network Merchants Offering and supersedes all prior oral and written understandings and representations relating to the subject matter of this Agreement.

2.3 Neither party shall be entitled to rely on any agreement, understanding or arrangement that is not expressly stated in this Agreement.

2.4 Nothing in this Agreement shall operate to limit or exclude liability for fraud.

2.5 This Agreement may not be modified except by written agreement between the parties.

3 Terms

3.1 Network Merchants maintains the Network Merchants Offering as a service to You in return for your evaluation and possible integration of the Site Content into your hardware and/or software applications. By accessing and/or using any part of the Network Merchants Offering, you are agreeing to comply with and be bound by the Terms. The Terms govern your access to and/or use of the Network Merchants Offering and any information, products, software or features made available to you via the Network Merchants Offering.

3.2 If you are using the Network Merchants Offering on behalf of your employer, you represent that you are authorised to accept these Terms on your employer's behalf.

3.3 In the case of any violation of the Terms, Network Merchants reserves the right to seek all remedies available by contract, law and in equity for such violations.

3.4 The Terms apply to all visits to and/or use of the Network Merchants Offering, both now and in the future.

3.5 ANY VIOLATION OF THE TERMS MAY RESULT IN THE SUSPENSION OR TERMINATION OF ACCESS TO AND/OR USE OF THE NETWORK MERCHANTS OFFERING OR SUCH OTHER ACTION PERMITTED IN ACCORDANCE WITH THE TERMS.

4 Website

4.1 You agree that access to and/or use of the Network Merchants Offering, is subject to all applicable national, international, state, and local laws and any and all applicable regulations, including the Terms.

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4.2 Network Merchants reserves the right to amend the Terms and modify or discontinue the Network Merchants Offering, temporarily or permanently, with or without notice, and is not obligated to support or update the Network Merchants Offering.

4.3 Terms amended under Clause 4.2 shall automatically become effective immediately after they are initially posted on the Site. Your continued access to and/or use of the Site after the posting of the amended Terms on the Site constitutes your affirmative acknowledgement of the Terms and the modifications and your agreement to abide and be bound by the Terms, as amended.

4.4 Should you object to any modifications of the Terms, your only recourse is to immediately terminate use of the Network Merchants Offering and notify Network Merchants of termination.

4.5 Portions of the Site may be password protected and/or secure areas. Access to such areas is restricted to authorized users only and unauthorised access or attempted access to these areas may be subject to prosecution.

5 Network Merchants IPR

5.1 Subject to the Terms, Network Merchants grants to You, a world-wide, non-transferable and non-exclusive right and licence for the duration of this Agreement to:

5.1.1 use the Network Merchants IPR and Program Concepts to develop and manufacture among others hardware and software applications which incorporate or access the Network Merchants IPR and Program Concepts;

5.1.2 use the Support Materials, but only as required to exercise the licence granted in 5.1.1 above.

5.2 Subject to the Terms, Network Merchants grants to You the right to use and modify the Support Materials supplied for the sole purpose of designing, developing and testing your software products, and to reproduce the Support Materials. Any use of the Network Merchants Offering other than as authorised under this Agreement is prohibited.

5.3 The following stipulations regarding the licence shall apply:

5.3.1 You shall prohibit any copying of the Network Merchants IPR, Support Materials, or Program Concepts by any third party.

5.3.2 on all copies made by You there shall be included proprietary, copyright, trade secret and warning legends, in the same form and location as any legend appearing on the original from which the copies are made, or in any other form and location specified by Network Merchants from time to time in writing.

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5.3.3 You are not permitted to decompile or reverse engineer the Network Merchants IPR and/or the Support Materials except to the extend permitted by statutory UK law.

6 Termination

6.1 Network Merchants may, in its sole discretion, immediately terminate or suspend this Agreement and/or access to or use of the Network Merchants Offering, with or without cause, at any time and with immediate effect. Network Merchants shall use reasonable endeavours to inform You of any such termination or suspension by written or electronic notice.

6.2 Upon termination of this Agreement and/or the Network Merchants Offering:

6.2.1 your right to access and/or use the Network Merchants Offering shall immediately cease; and

6.2.2 your rights under Clause 5 shall immediately cease; and

6.2.3 all other provisions of this Agreement capable of surviving termination shall continue in full force and effect

6.3 Network Merchants reserves the right to report any activity or disclose appropriate data or personal information that it suspects may violate any law or regulation to appropriate law enforcement officials, regulators, or other appropriate third parties. Network Merchants also may cooperate with appropriate law enforcement agencies to assist in the investigation and prosecution of any illegal conduct.

7 Proprietary Rights

7.1 The intellectual property in the Network Merchants Offering shall remain exclusively owned by Network Merchants. Nothing in this Agreement shall give You any right, title or interest in the intellectual property, save as expressly set out in this Agreement.

7.2 This Agreement shall not operate as an assignment to either party of any intellectual property right belonging or licensed, to the other party and each party shall retain ownership of, or other interest in, any such intellectual property right to which they may be entitled.

7.3 Nothing in this Agreement shall entitle either party to use the other party's name, logo, trademark or intellectual property right in any way or any context whatsoever without the prior written consent of the other.

7.4 You shall not register or attempt to register any marks or trademarks that would reasonably be deemed to be confusingly similar to any of Network Merchants's marks or trademarks.

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7.5 The Network Merchants Offering is proprietary to Network Merchants or to third parties from whom Network Merchants has obtained permission for use. Network Merchants authorises You to view, download, and print any part of the Network Merchants Offering provided that You:

7.5.1 comply with the Terms; and

7.5.2 do not publish or redistribute any Site Content except where agreed in writing by Network Merchants; and

7.5.3 do not modify the Site Content; and

7.5.4 do not remove any copyright, trademark, or other proprietary notices that have been placed on the Site or in the Site Content by Network Merchants. Except as expressly permitted above, reproduction or redistribution of the Site Content, or any portion of the Site Content, is strictly prohibited without the prior written permission of Network Merchants.

7.6 Network Merchants reserves the right to assign or sub-contract any of its rights or obligations under this Agreement.

7.7 You may not assign or sub-contract any of your rights or obligations under this Agreement.

7.7 Both parties will use their best endeavours to ensure that they do nothing that will bring disrepute to the other party's name or which will affect the other party's reputation.

8 Confidentiality

8.1 You shall not, without Network Merchants's prior written consent or as otherwise contemplated herein, disclose, provide, or make available any of the Site Content in any form to any person, except to employees, agents or consultants engaged by You whose access is necessary to enable You to exercise your rights under this Agreement.

8.2 For the avoidance of doubt Clause 8.1 shall remain in full force and effect whilst this Agreement is in force and for a period of 2 years from the date of termination of this Agreement howsoever caused.

8.3 The provisions set out in this Clause 8 are in addition to (and not in substitution for) all other confidentiality obligations between Network Merchants and You.

9 Indemnification

9.1 You agree to indemnify, defend and hold harmless Network Merchants, its parents, subsidiaries, affiliates, directors, officers, shareholders, agents, contractors and

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employees, from any claim or demand, including reasonable legal fees and court costs, made by any third party due to, arising from or out of:

9.1.1 your access to and/or use of the Network Merchants Offering; or

9.1.2 your breach or alleged breach of any representation, warranty or other obligation; or

9.1.3 your violation or alleged violation of any federal, state, international or local law and any and all regulations, rules or ordinances; or

9.1.4 the negligence or wilful misconduct by You or your employees or agents; or

9.1.5 your violation of the Terms;

9.1.6 your infringement of any intellectual property or other legal right of any person or entity.

10 Limitation of Liability

10.1 NETWORK MERCHANTS EXPRESSLY DISCLAIMS ANY LIABILITY OR LOSS ARISING FROM OR RELATED TO THE NETWORK MERCHANTS OFFERING (HOWEVER ARISING, INCLUDING NEGLIGENCE), INCLUDING WITHOUT LIMITATION, LIABILITY OR LOSS ASSOCIATED WITH UNAUTHORIZED ACCESS TO ANY SERVER, MERCHANT INTERFACES, WEBSITES, FACILITIES, YOUR DATA (INCLUDING CREDIT CARD NUMBERS AND OTHER PERSONALLY IDENTIFIABLE INFORMATION) WHETHER DUE TO ACCIDENT, ILLEGAL OR FRAUDULENT MEANS (INCLUDING HACKING, OR DEVICES USED BY ANY THIRD PARTY OR OTHER CAUSES BEYOND NETWORK MERCHANTS'S REASONABLE CONTROL).

10.2 YOU EXPRESSLY AGREE THAT NETWORK MERCHANTS SHALL NOT BE LIABLE FOR ANY LOSS ARISING FROM INFILTRATION OF THE NETWORK MERCHANTS OFFERING BY MEANS OF SOFTWARE VIRUSES, TROJAN HORSES, WORMS, TIME BOMBS, OR ANY OTHER SOFTWARE PROGRAMS, OR TECHNOLOGY DESIGNED OR INTENDED TO DISRUPT, DAMAGE, INTERCEPT, EXPROPRIATE, OBTAIN UNAUTHORISED ACCESS TO OR LIMIT THE FUNCTIONING OF THE NETWORK MERCHANTS OFFERING, YOUR DATA OR YOUR PERSONAL INFORMATION,

10.3 AS A CONDITION OF ACCESS TO AND/OR USE OF THE NETWORK MERCHANTS OFFERING, YOU AGREE THAT NEITHER NETWORK MERCHANTS, NOR ANY OFFICER, AFFILIATE, DIRECTOR, SHAREHOLDER, AGENT, CONTRACTOR OR EMPLOYEE OF NETWORK MERCHANTS WILL BE LIABLE TO YOU OR ANY THIRD PARTY FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, PUNITIVE, OR CONSEQUENTIAL LOSS OF PROFITS, LOSS OF EARNINGS, LOSS OF BUSINESS OPPORTUNITIES, DAMAGES, EXPENSES, OR COSTS RESULTING DIRECTLY OR INDIRECTLY FROM, OR OTHERWISE ARISING IN CONNECTION WITH THE

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NETWORK MERCHANTS OFFERING, HOWEVER ARISING, INCLUDING WITHOUT LIMITATION:

10.3.1 THE USE OF OR THE INABILITY TO USE THE NETWORK MERCHANTS OFFERING; AND/OR

10.3.2 THE COST OF PROCUREMENT OF SUBSTITUTE GOODS AND SERVICES; AND/OR

10.3.3 ANY DATA ENTERED BY YOU THROUGH THE SITE; AND/OR

10.3.4 RESULTING FROM UNAUTHORIZED ACCESS TO, DELETION OR ALTERATION OF YOUR DATA; AND/OR

10.3.5 RESULTING FROM ANY FAILURE OF THE SITE TO STORE YOUR DATA.

THIS CLAUSE 10.3 WILL APPLY EVEN IF NETWORK MERCHANTS HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

10.4 THE LIMITATIONS SET FORTH IN THIS SECTION APPLY TO THE ACTS, OMISSIONS, NEGLIGENCE, AND GROSS NEGLIGENCE OF NETWORK MERCHANTS AND ITS AFFILIATES, PARENTS, SUBSIDIARIES, CONTRACTORS, SUBCONTRACTORS, OFFICERS, DIRECTORS, SHAREHOLDERS, MANAGERS, EMPLOYEES, AND AGENTS, WHICH, BUT FOR THIS PROVISION, WOULD GIVE RISE TO A COURSE OF ACTION IN CONTRACT, OR ANY OTHER LEGAL DOCTRINE.

10.5 EXCEPT AS OTHERWISE LIMITED, NETWORK MERCHANTS'S LIABILITY AND THE LIABILITY OF ANY SUBSIDIARIES, OFFICERS, DIRECTORS, EMPLOYEES AND AGENTS TO YOU OR ANY THIRD PARTIES IN ANY CIRCUMSTANCES SHALL BE LIMITED TO THE LESSER OF:

10.5.1 THE AMOUNT YOU PAY US IN THE 30 DAYS PRIOR TO THE ACTION GIVING RISE TO LIABILITY; OR

10.5.2 £25,000.

10.6 NETWORK MERCHANTS SHALL NOT BE LIABLE TO YOU OR ANY THIRD PARTY FOR TERMINATION OF THIS AGREEMENT.

10.7 YOU ARE SOLELY RESPONSIBLE FOR IDENTIFYING AND OBTAINING ANY AND ALL PATENT OR OTHER INTELLECTUAL PROPERTY LICENCES THAT MAY BE NEEDED FOR YOUR OWN HARDWARE AND SOFTWARE PRODUCTS.

10.8 EXCEPT WHERE OTHERWISE EXPRESSLY AGREED IN WRITING BETWEEN NETWORK MERCHANTS AND YOU, THE NETWORK MERCHANTS OFFERING IS PROVIDED IN STANDARD FORMAT, AND NETWORK MERCHANTS HAS NOT BEEN ENGAGED TO DEVELOP THE NETWORK MERCHANTS OFFERING FOR USE WITH YOUR HARDWARE OR SOFTWARE PRODUCTS. AS A RESULT, IT IS YOUR SOLE

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RESPONSIBILITY TO DETERMINE THE SUITABILITY OF THE NETWORK MERCHANTS OFFERING FOR ANY INTENDED USE OR PURPOSE.

11 Warranties

12.1 THE NETWORK MERCHANTS OFFERING IS PROVIDED ON AN "AS IS," "AS AVAILABLE" BASIS. NETWORK MERCHANTS DOES NOT REPRESENT OR WARRANT THAT THE NETWORK MERCHANTS OFFERING WILL BE AVAILABLE, ACCESSIBLE, UNINTERRUPTED, TIMELY, SECURE, ACCURATE, COMPLETE, ENTIRELY ERROR-FREE OR VIRUS FREE. NETWORK MERCHANTS DOES NOT MAKE ANY WARRANTY AS TO THE RESULTS THAT MAY BE OBTAINED FROM THE USE OF THE NETWORK MERCHANTS OFFERING OR AS TO THE ACCURACY OR RELIABILITY OF ANY CONTENT OR ANY INFORMATION OR PRODUCTS OBTAINED VIA THE NETWORK MERCHANTS OFFERING. NETWORK MERCHANTS DOES NOT MAKE ANY WARRANTY THAT DEFECTS IN THE SERVICES WILL BE CORRECTED. YOU EXPRESSLY ACKNOWLEDGE THAT THE NETWORK MERCHANTS OFFERING IS A COMPUTER NETWORK-BASED SERVICE, WHICH MAY BE SUBJECT TO OUTAGES, INTERRUPTIONS, ATTACKS BY THIRD PARTIES AND DELAY OCCURRENCES. IN SUCH AN EVENT AND SUBJECT TO THE TERMS HEREOF, NETWORK MERCHANTS SHALL USE COMMERCIALLY REASONABLE EFFORTS TO REMEDY MATERIAL INTERRUPTIONS AND WILL PROVIDE ADJUSTMENTS, REPAIRS AND REPLACEMENTS, WITHIN ITS CAPACITY, THAT ARE NECESSARY TO ENABLE THE NETWORK MERCHANTS OFFERING TO PERFORM ITS INTENDED FUNCTIONS IN A REASONABLE MANNER. YOU ACKNOWLEDGE THAT NETWORK MERCHANTS DOES NOT WARRANT THAT SUCH EFFORTS WILL BE SUCCESSFUL. IF NETWORK MERCHANTS'S EFFORTS ARE NOT SUCCESSFUL, YOU MAY TERMINATE THE APPLICABLE SERVICE AGREEMENT AND CEASE TO ACCESS AND/OR USE THE NETWORK MERCHANTS OFFERING.

11.2 THE FOREGOING SHALL CONSTITUTE YOUR SOLE REMEDY, AND NETWORK MERCHANTS'S SOLE LIABILITY, IN THE EVENT OF INTERRUPTION, OUTAGE OR OTHER DELAY OCCURRENCES.

11.3 YOU WAIVE AND COVENANT NOT TO ASSERT ANY CLAIMS OR ALLEGATIONS OF ANY NATURE ARISING FROM OR RELATING TO YOUR USE OF THE NETWORK MERCHANTS OFFERING, INCLUDING, WITHOUT LIMITATION, ALL CLAIMS AND ALLEGATIONS RELATING TO THE ALLEGED INFRINGEMENT OF PROPRIETARY RIGHTS, THE ALLEGED INACCURACY OF CONTENT, OR ALLEGATIONS THAT NETWORK MERCHANTS HAS OR SHOULD INDEMNIFY, DEFEND, OR HOLD YOU HARMLESS FROM ANY CLAIM OR ALLEGATION ARISING FROM YOUR ACCESS TO AND/OR USE OF THE NETWORK MERCHANTS OFFERING.

11.4 YOU ACCESS AND/OR USE THE NETWORK MERCHANTS OFFERING AT YOUR OWN RISK AND NEITHER NETWORK MERCHANTS NOR THE PARTIES WHO

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PROVIDE INFORMATION AND MATERIALS FOR PUBLICATION WITHIN THE SITE WILL BE LIABLE FOR ANY DAMAGES ARISING FROM YOUR USE OF THIS SITE.

11.5 YOU UNDERSTAND AND AGREE THAT ANY MATERIAL OR DATA DOWNLOADED OR OTHERWISE OBTAINED THROUGH THE ACCESS TO AND/OR USE OF THE NETWORK MERCHANTS OFFERING IS DONE AT YOUR OWN DISCRETION AND RISK AND THAT YOU WILL BE SOLELY RESPONSIBLE FOR ANY DAMAGE TO YOUR COMPUTER SYSTEM, LOSS OF SERVICE OR LOSS OF DATA THAT RESULTS FROM THE DOWNLOAD OR USE OF SUCH MATERIAL OR DATA.

11.6 EXCEPT AS EXPRESSLY SET FORTH IN THIS CLAUSE 12, NETWORK MERCHANTS SPECIFICALLY DISCLAIMS ALL REPRESENTATIONS, WARRANTIES, AND CONDITIONS WHETHER EXPRESS OR IMPLIED, ARISING BY STATUTE, OPERATION OF LAW, USAGE OF TRADE, COURSE OF DEALING, OR OTHERWISE, INCLUDING BUT NOT LIMITED TO, WARRANTIES OR CONDITIONS OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NON-INFRINGEMENT, OR TITLE WITH RESPECT TO THE NETWORK MERCHANTS OFFERING.

11.7 YOU AGREE THAT NETWORK MERCHANTS DOES NOT REPRESENT OR WARRANT THAT THE NETWORK MERCHANTS OFFERING WILL BE AVAILABLE, ACCESSIBLE, UNINTERRUPTED, TIMELY, SECURE, ACCURATE, COMPLETE, OR ENTIRELY ERROR-FREE.

11.8 YOU REPRESENT AND WARRANT THAT YOUR ACCESS TO AND USE OF THE NETWORK MERCHANTS OFFERING WILL BE CONSISTENT WITH THE TERMS AND WILL NOT INFRINGE OR VIOLATE THE RIGHTS OF ANY OTHER PARTY OR BREACH ANY CONTRACT OR LEGAL DUTY TO ANY OTHER PARTIES.

11.9 YOU SHALL BE SOLELY RESPONSIBLE FOR ENSURING:

11.9.1 THE ACCURACY OF ANY INPUT DATA, AND

11.9.2 PROPER USE OF THE NETWORK MERCHANTS OFFERING.

12 Privacy Policy

Information collected by Network Merchants via the Site will be treated in accordance with Network Merchants's Privacy Policy which may be found on the Site and which is incorporated herein by reference.

13 Force Majeure

13.1 Network Merchants shall not be liable in respect of any breach of this Agreement due to any cause beyond its reasonable control including but not limited to Act of God, flood, lightning or fire, the act or omission of Government, war, terrorism, riot, strikes or other labour disputes, fires, transportation contingencies, congestion or non-operation of

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any service or product provided by a third party, acts or omissions by a third party, infiltration or disruption of the Network Merchants Offering.

14 Waiver

14.1 No failure or delay by either party in exercising its rights under this Agreement shall be construed as a waiver or release from that right or create a precedent or in any way prejudice any party's rights under this Agreement.

14.2 Any waiver by either party of any provision of this Agreement shall be limited to the particular breach and shall not operate in any way in respect of any future breach.

14.3 The rights and remedies provided in this Agreement are cumulative and are additional to any rights or remedies provided by law.

15 Invalidity Clause

15.1 If any provision of this Agreement is held by any court or other competent authority to be invalid or unenforceable in whole or in part:-

15.1.1 all other provisions of the Agreement not affected by such invalidity or unenforceability shall remain in full force and effect; and

15.1.2 both parties will seek to agree a valid and enforceable substitute provision that achieves to the greatest extent possible the economic legal and commercial objective of the invalid or unenforceable provision.

16 Agency

16.1 Nothing in this Agreement shall be deemed to constitute any partnership between the parties or to create any relationship of agency between them and neither of the parties has authority to bind the other party unless expressly stated in this Agreement.

17 Third Party Rights

17.1 The parties do not intend any third party (other than as expressly provided for herein) to have the right to enforce any provision of this Agreement under the Contracts (Rights of Third Parties) Act 1999 or otherwise. Any rights conferred on any third party hereunder may be altered or extinguished by the written agreement between the parties without the consent of the third party.

18 Law

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18.1 This Agreement and the performance of all obligations hereunder shall be governed by and construed in accordance with English Law and the parties hereto hereby submit to the exclusive jurisdiction of the English Courts.

18.2 You consent to the personal jurisdiction of the English Courts over any claim arising from this Agreement, stipulate to the fairness and convenience of proceeding in such courts, and covenant not to assert any objections to proceeding in such courts.

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